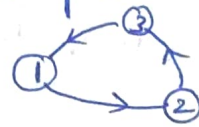


Irreducible & Ergodic

- 1) State 1 $\rightarrow$ state 2
State 2 $\rightarrow$ state 3.
State 3 $\rightarrow$ state 1

$\rightarrow$ This chain is irreducible because you can reach any state from any other state in a finite no. of steps. For example

$$1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 1$$



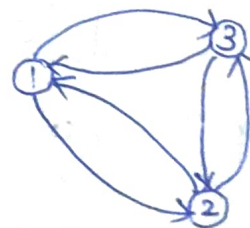
$\rightarrow$ Here the chain is periodic. If you start in state 1, you can only return to it in

3 steps, 6 steps, 9 steps etc. Here period is 3.

$\rightarrow$ Since it is periodic, the chain is not ergodic.

- 2) State $\leftrightarrow$ state 2 $\leftrightarrow$ state 3

$\rightarrow$ This chain is irreducible because you can go from any ~~old~~ state to any other state.



$\rightarrow$ It is aperiodic because there is no fixed no. of steps reqd to return to any state.

For example $\text{Prob}(1 \rightarrow 1)$ in one step might be non-zero (~~not~~ self loop), breaking periodicity.

→ All the states are positive recurrent, means that you will revisit them eventually on average in a finite expected time.

→ Hence the chain is ergodic.

→ A positive recurrent (positive persistent) and aperiodic state is called ergodic. A Markov chain of all those states is called ergodic chain.

→ The prob. that the chain starting from state i returns to state i for the first time at the n^{th} step is denoted by $f_{ii}^{(n)}$, $n=1,2,3,\dots$ and is called first time return time prob. or recurrence time prob.

→ If $F_{ii} = \sum_{n=1}^{\infty} f_{ii}^{(n)} = 1$, the return to state i is certain and the state i is said to be persistent or recurrent.

→ If $F_{ii} < 1$ (return to state i is uncertain) then state i is said to be transient.

→ The parameter $\mu_{ii} = \sum_{n=1}^{\infty} n f_{ii}^{(n)}$ is called mean recurrence time of state i .

(a) If μ_{ii} is finite then i is said to be non-null persistent or +ve persistent.

(b) If $\mu_{ii} = \infty$ then i is said to be null persistent.

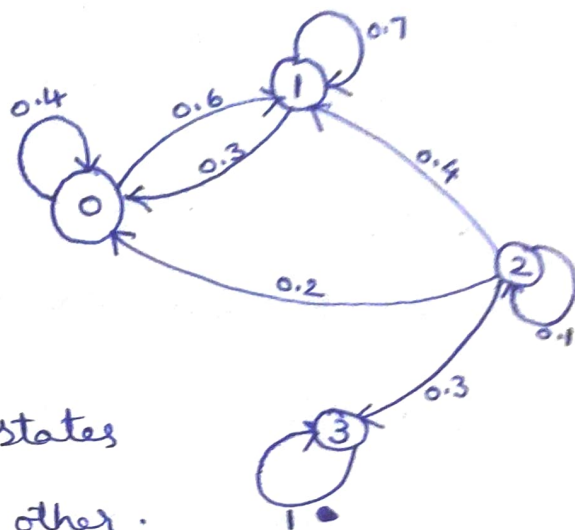
1. Consider the markov chain with transition probability

matrix
$$\begin{bmatrix} 0.4 & 0.6 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Is this matrix irreducible?

Soln.

$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.4 & 0.6 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$



It is clear that all the states donot communicate with each other. From state 3, no other state is accessible. State 3 is an absorbing state. So, the chain is not irreducible or ergodic.

2. check whether the following markov chain is regular & ergodic.

Soln.

$$P = \begin{bmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 1/2 & 1/2 & 0 \end{bmatrix}$$

$$\begin{aligned} P^2 &= \begin{bmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 1/2 & 1/2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & 0 & 0 & 1/2 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \end{bmatrix} \end{aligned}$$

$$\text{Now } p^3 = p^2 \cdot p = \begin{bmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} = p.$$

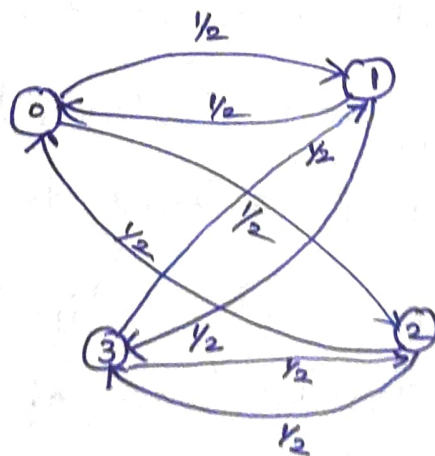
$$p^4 = p^3 \cdot p = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} = p^2$$

.....

$$p^3 = p^5 = p^7 = \dots = p$$

$$p^2 = p^4 = p^6 = \dots$$

As not all the elements in any matrices are non-zeros, given matrix p is not regular.



It is possible to go from state 0 to state 1 or state 2, from state 1 to state 0 or state 2, from state 2 to state 0 or state 3, from state 3 to state 1 or state 2. Hence all the states are ergodic & chain is ergodic chain.

3) Find the nature of states of the Markov chain with t.p.m.

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

Sol

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

$$P^3 = P^2 \cdot P = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix} = P$$

$$P^4 = P^3 \cdot P = P^2$$

$$\therefore P^3 = P^5 = P^7 = \dots = P$$

$$P^4 = P^6 = \dots = P^2$$

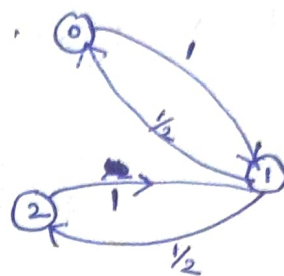
from the above, $P_{00}^{(2)}, P_{11}^{(2)}, P_{22}^{(2)}$ are all > 0
 $P_{00}^{(4)}, P_{11}^{(4)}, P_{22}^{(4)}$ are all > 0

$$\therefore d_i = \text{G.C.D} \{ 2, 4, 6, \dots \}$$

$$= 2$$

$$\therefore d_i = \text{G.C.D} \{ m, P_{ii}^{(m)} > 0 \}$$

$\therefore d_i > 1$, States are periodic with period 2.



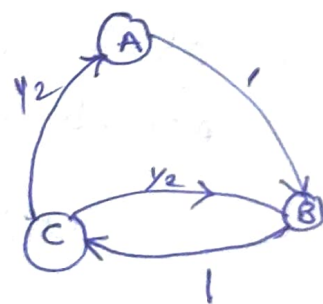
clearly as all the states are able to communicate given matrix is irreducible.

It is not ergodic as states of Markov chain are not aperiodic.

4). Three boys A, B, & C are throwing a ball to each other. A always throws the ball to B & B always throws the ball to C; but C is just as likely to throw the ball to B as to A. Show that the process is Markovian. Find the transition matrix & classify the states.

Sol . The transition ~~matrix~~ probability matrix of the process $\{X_n\}$ is given as.

$$P = \begin{matrix} & \begin{matrix} \text{states of } X_n \\ \text{A} & \text{B} & \text{C} \end{matrix} \\ \begin{matrix} \text{states of } X_{n-1} \\ \text{A} \\ \text{B} \\ \text{C} \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \end{matrix}$$



Since states of X_n depend only on states of X_{n-1} but not on states of $X_{n-2}, X_{n-3}, \dots$
 $\{X_n\}$ is a Markov chain.

$$P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix},$$

$$P^3 = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}, \quad P^4 = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 1/8 & 3/8 & 1/2 \end{bmatrix}$$

$$P^6 = \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 3/8 & 1/2 \\ 1/8 & 3/8 & 3/8 \end{bmatrix}$$

Here $P_{22}^{(2)} > 0, P_{33}^{(2)} > 0, P_{22}^{(3)} > 0, P_{33}^{(3)} > 0$

$\therefore d_i = \text{G.C.D} \{ 2, 3, 4, \dots \} = 1$ for states 2 & 3

By $P_{11}^{(3)} > 0, P_{11}^{(4)} > 0, \dots$

$\therefore d_i = \text{G.C.D} \{ 3, 4, \dots \} = 1$ for state 1.

Hence all the states are aperiodic.

→ chain is finite & irreducible.

All its states are non-null persistent as

$u_{ii} = \sum n f_{ii}^{(n)} = \text{finite}$

$n \times f_{11} = 3 \times \frac{1}{2} + 5 \times \frac{1}{4} + 6 \times \frac{1}{4} = \text{finite}$

All the states are ergodic.

$n \times f_{22} = 2 \times \frac{1}{2} + 3 \times \frac{1}{2} + 4 \times \frac{1}{2} + 5 \times \frac{1}{2} + 6 \times \frac{3}{8} = \text{finite}$

$n \times f_{33} = 2 \times \frac{1}{2} + 3 \times \frac{1}{2} + 4 \times \frac{1}{4} + \dots = \text{finite}$

A special chain of ergodic chain is regular chain. Here P is regular chain clearly from P⁵.